

Jacob Zhang

Embedded Systems & IoT Firmware Engineer • 3 Years Experience • MERIQ Index: 87% (Strong Candidate)

■ jacob.zhang159@example.com | ■ +1 (368) 984-9693 | ■ Dublin, Ireland

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Embedded Linux, I2C/SPI/UART, PCB Debugging, FreeRTOS, ARM Cortex, MQTT, C/C++

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Embedded Linux service, reducing latency by 26%.
- Led a squad of 4 engineers delivering scalable microservices processing 39M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 31%.

Associate Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - 2023)

- Designed and maintained I2C/SPI/UART data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using PCB Debugging saving \$11k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [Embedded Linux, I2C/SPI/UART, PCB Debugging, FreeRTOS]

Engineered end-to-end production solution leveraging Embedded Linux and I2C/SPI/UART with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [PCB Debugging, FreeRTOS, ARM Cortex, MQTT]

Architected high-resilience system integrating PCB Debugging with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

Imperial College London — 2019 - 2023 (Score: 9.66 / 10.0)